NAAN MUDHALVAN ARTS INTERNSHIP PROGRAM 2026
PYTHON FOR DATA ANALYTICS
PROJECT SUBMISSION TEMPLATE

Hospital Patient Data Analysis

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TITLE	: Hospital Patient Data Analysis
GITHUB LINK	:
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1. ABSTRACT

This project focuses on analyzing hospital patient records to improve healthcare service efficiency and identify common diseases affecting patients. A hospital dataset containing patient details such as age, disease, admission date, discharge date, and treatment information is analyzed using Python Data Analytics techniques. Data cleaning is performed to handle missing values, remove duplicates, and ensure data consistency. Exploratory Data Analysis (EDA) is used to understand disease distribution, patient demographics, and admission patterns. Visualization techniques are applied to represent trends using charts and graphs. The project helps hospital management understand patient flow, identify peak admission periods, and make informed decisions regarding resource allocation and healthcare planning.

2. PROBLEM STATEMENT

Hospitals generate large amounts of patient data every day. Manual analysis of these records is difficult and time-consuming. Without proper analysis, hospitals may face challenges in resource planning, disease monitoring, and patient management. Data analytics helps transform raw patient data into meaningful insights that improve healthcare services.

3. OBJECTIVES

- Clean and preprocess hospital patient data.
- Analyze patient count by disease.
- Study patient distribution by age group.
- Identify peak admission periods.
- Visualize patient trends using charts.
- Generate useful insights for hospital management.

4. PROJECT SCOPE

The project focuses on analyzing hospital patient records using Python in Google Colab.

- The scope includes data cleaning,
- Exploratory data analysis
- Visualization
- Reporting

The analysis supports healthcare decision-making and resource management.

5. TECHNOLOGIES USED

Development Environment: Google Colab

Programming Language:
- Python

Libraries Used:
- Pandas
- NumPy
- Matplotlib
- Seaborn

Dataset Source:
- CSV File containing hospital patient records

6. APPLICATION FLOW

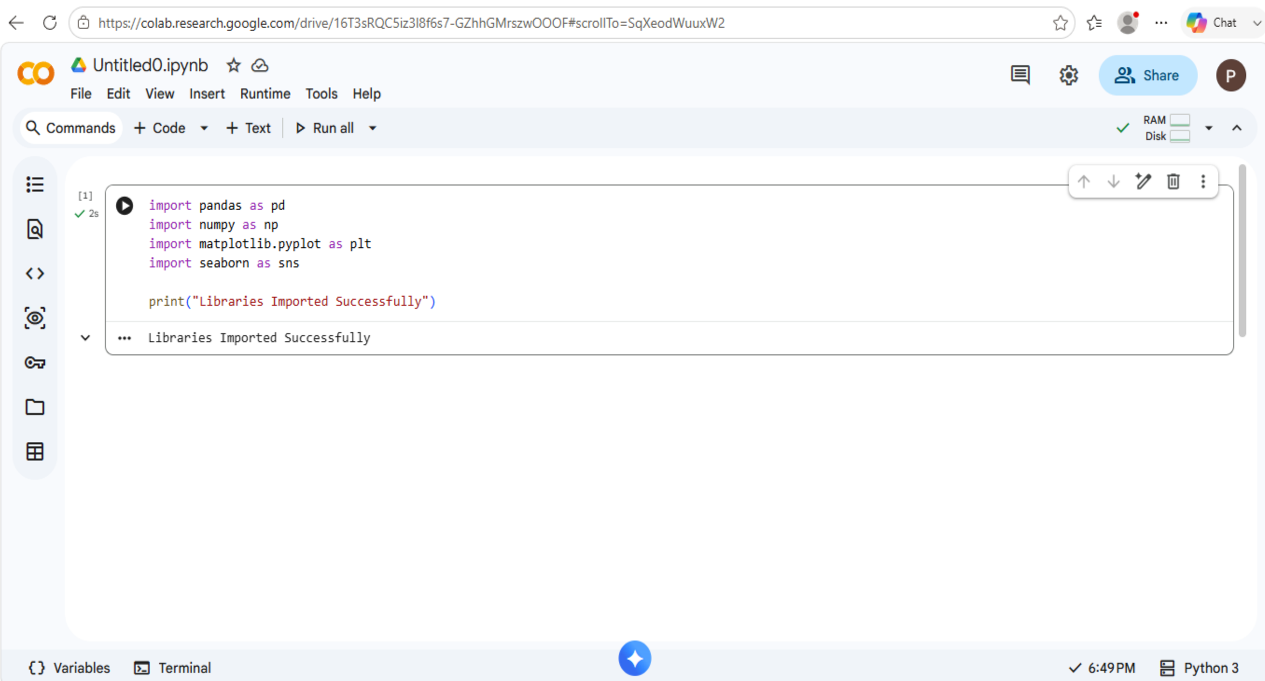
Step 1: Load Dataset
Step 2: Data Cleaning
Step 3: Handle Missing Values
Step 4: Remove Duplicate Records
Step 5: Disease Analysis
Step 6: Age Group Analysis
Step 7: Admission Trend Analysis
Step 8: Create Visualizations
Step 9: Generate Final Insights

7. KEY PYTHON DATA ANALYTICS CONCEPTS USED

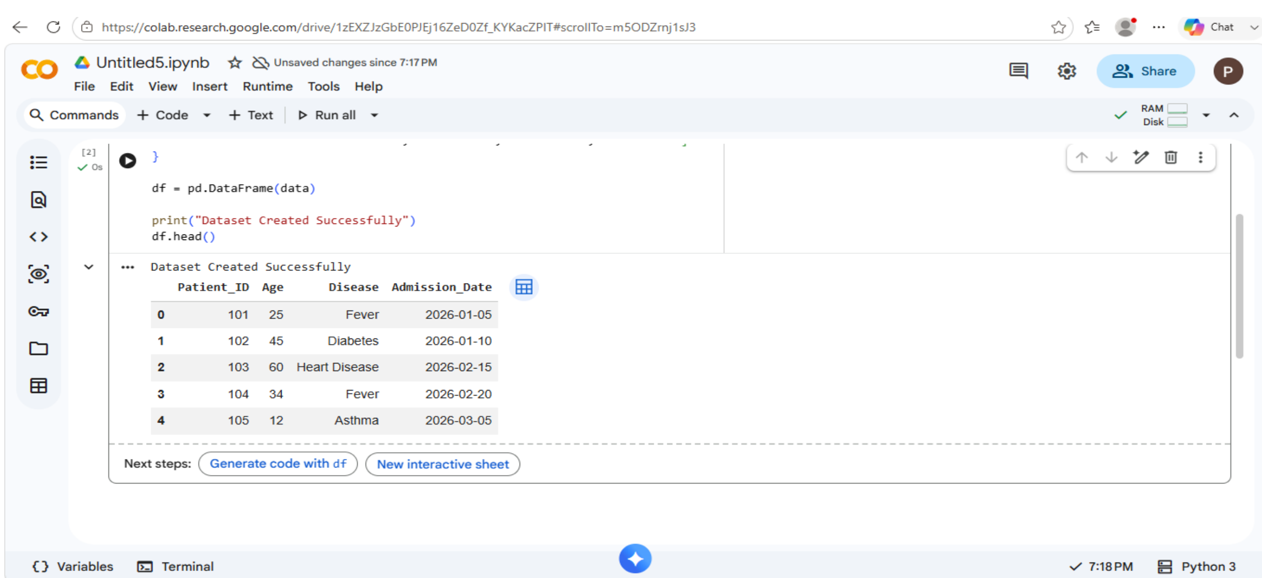
1. Data Handling using Pandas
2. Data Cleaning
3. Exploratory Data Analysis (EDA)
4. Data Visualization
5. Grouping and Aggregation
6. Trend Analysis

8. EXPECTED OUTPUT

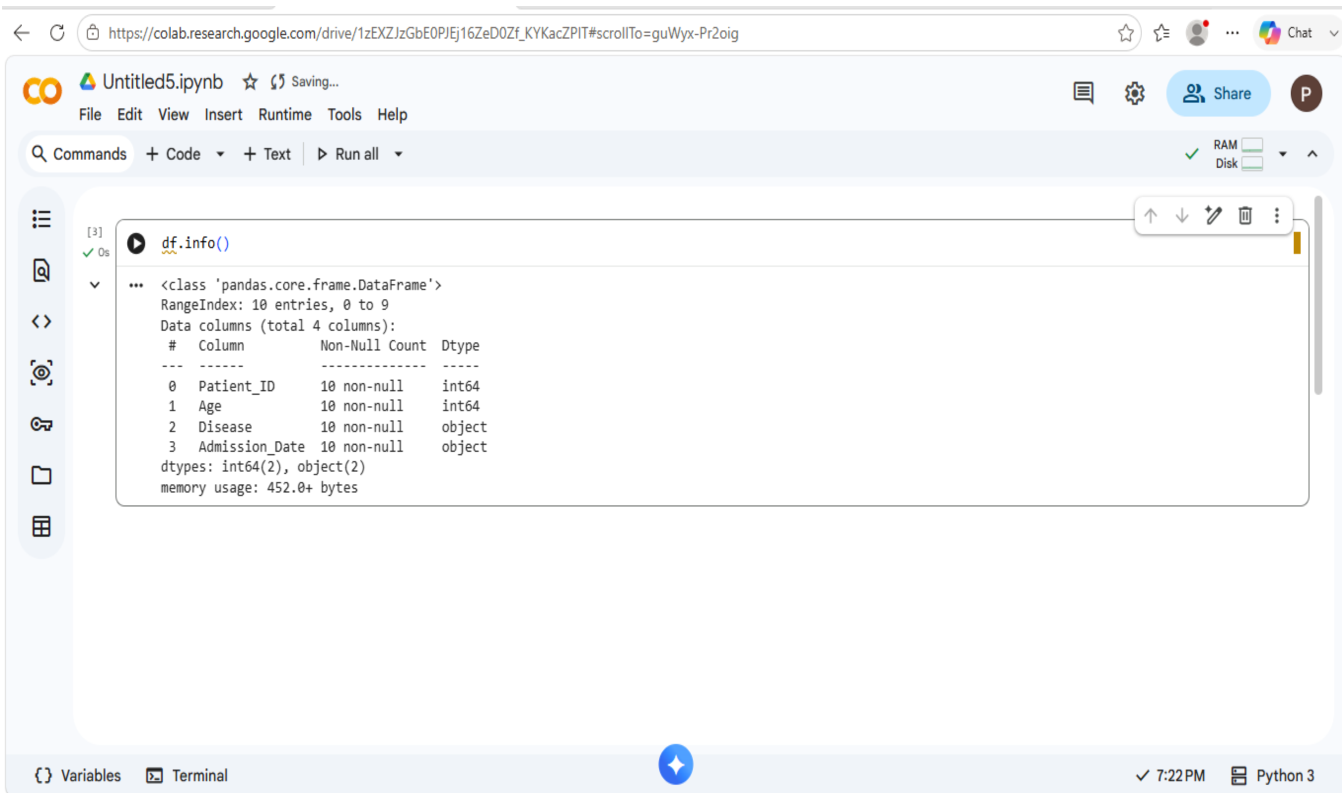
Dataset Preview



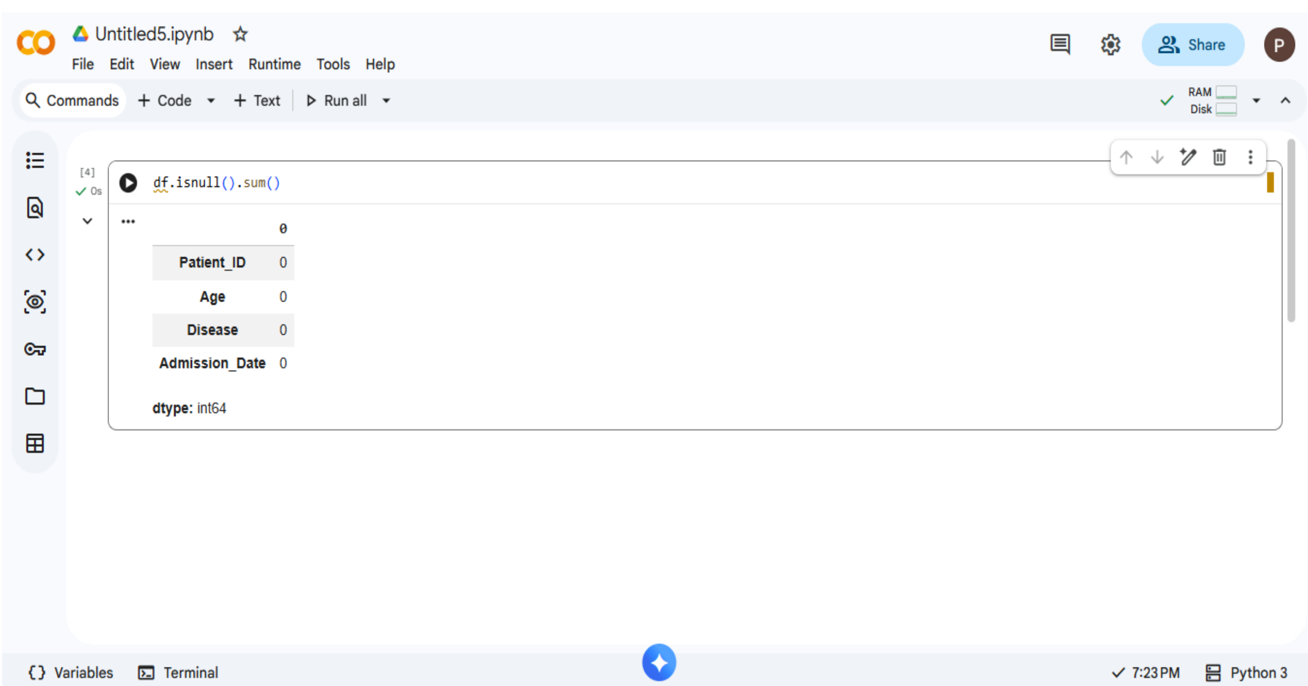
Dataset Information



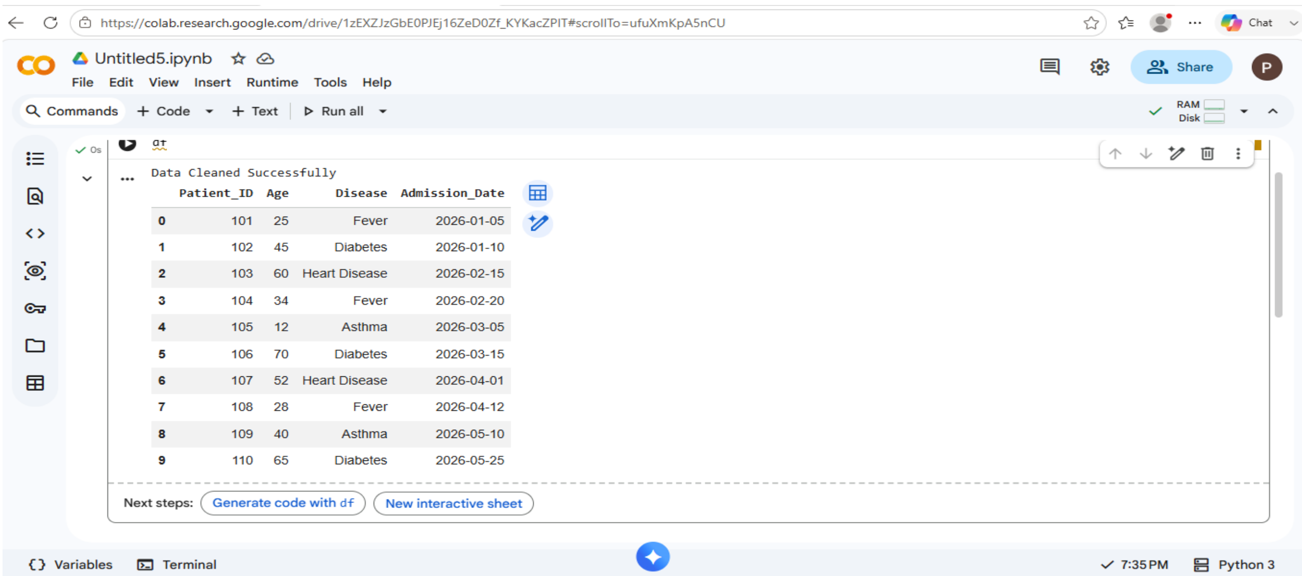
Missing Values Check



Cleaned Dataset

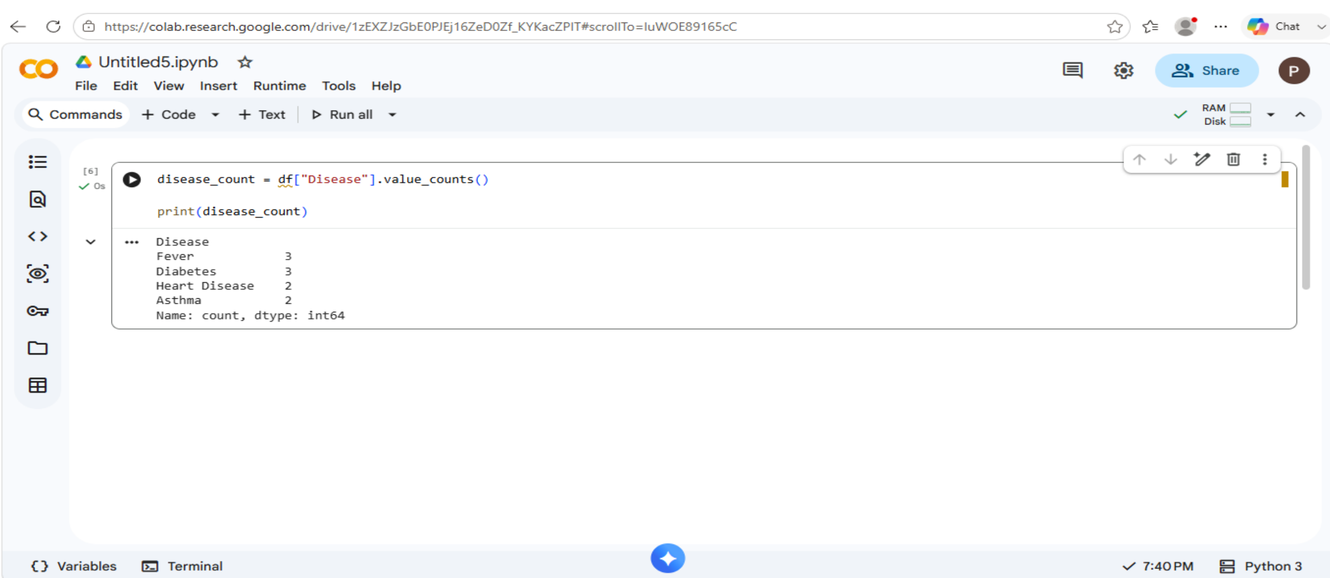


Disease Count Table



	Patient_ID	Age	Disease	Admission_Date
0	101	25	Fever	2026-01-05
1	102	45	Diabetes	2026-01-10
2	103	60	Heart Disease	2026-02-15
3	104	34	Fever	2026-02-20
4	105	12	Asthma	2026-03-05
5	106	70	Diabetes	2026-03-15
6	107	52	Heart Disease	2026-04-01
7	108	28	Fever	2026-04-12
8	109	40	Asthma	2026-05-10
9	110	65	Diabetes	2026-05-25

Disease Count Graph

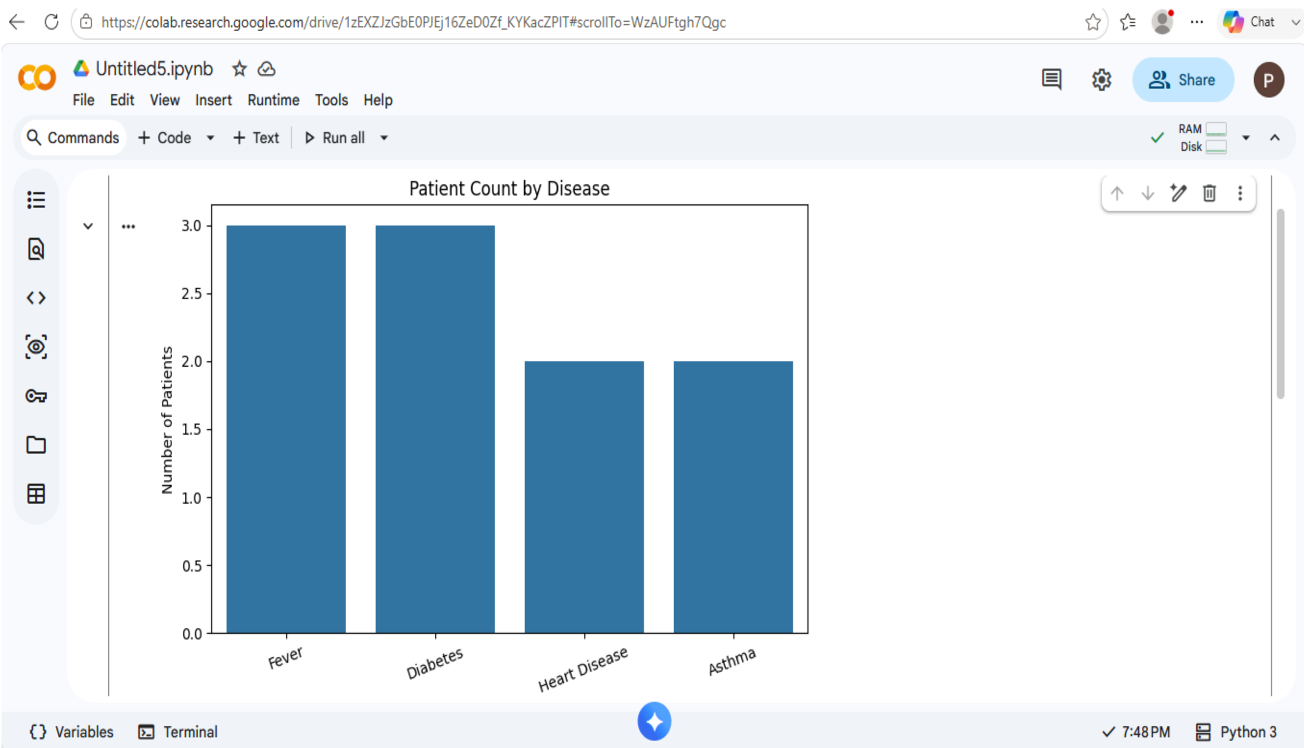



```
disease_count = df["Disease"].value_counts()

print(disease_count)
```

```
Disease
Fever            3
Diabetes         3
Heart Disease    2
Asthma           2
Name: count, dtype: int64
```

Age Group Dataset



Age Group Count

	Patient_ID	Age	Disease	Admission_Date	Age_Group
0	101	25	Fever	2026-01-05	Young Adult
1	102	45	Diabetes	2026-01-10	Adult
2	103	60	Heart Disease	2026-02-15	Senior
3	104	34	Fever	2026-02-20	Young Adult
4	105	12	Asthma	2026-03-05	Child
5	106	70	Diabetes	2026-03-15	Senior
6	107	52	Heart Disease	2026-04-01	Senior
7	108	28	Fever	2026-04-12	Young Adult
8	109	40	Asthma	2026-05-10	Adult
9	110	65	Diabetes	2026-05-25	Senior

Next steps: Generate code with df | New interactive sheet

Age Group Graph

https://colab.research.google.com/drive/1zEXZJzGbE0PJEj16ZeD0Zf_KYKacZPIT#scrollTo=8A6zOZE09sC3

```
age_group_count = df["Age_Group"].value_counts()

print(age_group_count)
```

```
Age_Group
Senior         4
Young Adult    3
Adult          2
Child          1
Name: count, dtype: int64
```

Monthly Admission Table

```
df["Admission_Date"] = pd.to_datetime(df["Admission_Date"])

df["Month"] = df["Admission_Date"].dt.month_name()

monthly_admission = df["Month"].value_counts()

print(monthly_admission)
```

```
Month
January     2
February    2
March       2
April       2
May         2
Name: count, dtype: int64
```

Monthly Admission Graph

Final Insights

```
print("\nMost Common Disease:")
print(df["Disease"].value_counts().idxmax())

print("\nTotal Patients:")
print(len(df))

print("\nPeak Admission Month:")
print(monthly_admission.idxmax())
```

```
FINAL INSIGHTS

Most Common Disease:
Fever

Total Patients:
10

Peak Admission Month:
January
```

9. CHALLENGES FACED

• Missing values in patient records
• Duplicate entries
• Incorrect date formats
• Data inconsistency issues
• Large dataset handling

10. LEARNING OUTCOMES

- Learned data cleaning techniques.
- Improved Python programming skills.
- Gained practical experience with Pandas and NumPy.
- Learned data visualization using Matplotlib and Seaborn.
- Developed analytical thinking and problem-solving skills.

11. CONCLUSION

The Hospital Patient Data Analysis project successfully analyzes patient records and provides meaningful insights. The project identifies common diseases, analyzes patient demographics, and determines peak admission periods. The generated insights can help hospitals improve resource allocation, patient care, and operational efficiency. This project demonstrates the importance of data analytics in the healthcare sector.